

OEE Dashboard

Manufacturing Intelligence System

Overall Equipment Effectiveness · Quality · Production · Issues

Excel

Replacing

Morning Huddle

FastAPI

Built With

+ React + SQLite

OEE

Tracking

Avail · Perf · Quality

The Problem

Every morning, production supervisors ran their huddle off a shared Excel spreadsheet.

No Real-Time Data ⚠️

Numbers were entered manually each morning — stale by the time anyone read them

Disconnected Issues 📋

Foremen reported problems verbally with no structured tracking or history

Manual OEE Calculation 📊

Availability, Performance, and Quality were calculated by hand in Excel every week

No Notifications 🔔

Supervisors had no way to know when new issues were submitted between huddles

The Solution

A full-stack web dashboard replacing Excel — accessible to every supervisor and foreman on the network.

Live Dashboard



Real-time data — no manual entry, 30-second auto-refresh

Issue Tracking



Structured issue submission, updates, and resolution history

OEE Calculations



Automatic Availability, Performance, Quality, and OEE metrics

Push Notifications



Browser notifications when new issues or work orders arrive

Morning Huddle



Guided 3-phase presentation mode for daily stand-up meetings

Excel Export



One-click export to xlsx — issues, work orders, labor hours

Tech Stack

Frontend

- React 18 + Vite
- Recharts (OEE graphs)
- xlsx (Excel export)
- react-helmet-async

Single-page app served via IIS on the network server

Backend

- Python 3.14 + FastAPI
- SQLAlchemy ORM
- Pydantic schemas
- Uvicorn ASGI server

REST API with 40+ endpoints, running as a Windows service

Database

- SQLite (oee.db)
- 12 tables
- Seeded truck & defect types
- Goal history tracking

Zero-config file-based database, deployed to network share

Infrastructure

- NSSM service manager
- IIS web server
- Network share (Z:\)
- GitHub version control

Deployed to \\RGSAPPSRV — accessible to all plant PCs

Database Schema

issues

id, issue_type, category, description, foreman_name, status, solved_by, is_read

issue_updates

id, issue_id, update_num, note, made_by, created_at

work_orders

id, work_order_num, truck_type, total_defects, week_start, is_read

work_order_defects

id, work_order_id, defect_type_id, quantity

defect_types

id, name – Sharp Edges, Missing Parts, ...

truck_types

id, name – Monmouth, PennDOT, Steel, Alum Comm

indirect_labor

id, week_start, working_days, total_labor_hours, indirect_hours, rework_hours

oeo_goals

annual_dpu_goal, quarterly_dpu_goal, weekly_trucks_min/max

goal_history

id, effective_date, annual_dpu_goal, quarterly_dpu_goal

foremen

id, name (unique)

supervisors

id, name (unique)

rework_hours

id, week_start, hours, notes

Issues Module

Foreman Workflow

- 1

Foreman visits /foreman and submits a part or process issue with category and description
- 2

Issue appears on supervisor dashboard with NEW badge and browser push notification
- 3

Supervisor adds updates, changes status to In Progress, assigns resolution
- 4

Issue marked Solved with resolution note and solved-by supervisor name

Supervisor Features

- LIVE

30-Second Auto-Refresh

Dashboard polls every 30s; new issues trigger desktop push notifications
- SORT

Grouped by Foreman

Issues organized by foreman in configurable order, collapsible sections
- BULK

Mass Add + Bulk Delete

Submit multiple issues at once; checkbox select and bulk delete
- HIST

Full Update History

Unlimited updates per issue with timestamp, author, and note

OEE Metrics

Three components multiply to produce the Overall Equipment Effectiveness score:

Availability

$$\frac{(\text{Total Hours} - \text{Indirect} - \text{Rework})}{\text{Total Hours}}$$

Measures how much of scheduled time was actually productive. Pulled from weekly labor hour entries.

Performance

$$\frac{\text{Actual Trucks}}{\text{Target Trucks per Week}}$$

Compares actual production to the weekly target (14–18 trucks). Adjusts for short weeks automatically.

Quality

$$\frac{\text{Quarterly DPU Goal}}{\text{Actual DPU}}$$

Defects per unit vs. the quarterly goal. Capped at 100% when under target.

$$\text{OEE} = \text{Availability} \times \text{Performance} \times \text{Quality}$$

Period filtering: Last Week · Month · Quarter · Full Year — with trend arrows comparing to prior period

Work Orders & Defect Tracking

Work Order Entry



Per-Truck Logging

Each work order records work order #, truck type, and defect breakdown by type



Defect Breakdown

Multiple defect types per work order (Sharp Edges, Missing Parts, etc.) with quantities



Foreman Form

Foremen submit directly at /defects — lightweight form, no login required



Unread Badges

New submissions show red NEW badges on supervisor dashboard with notifications

Analysis & History

Defect Type Charts

Pie + stacked bar charts showing defect distribution, filterable by period and truck type

Weekly DPU History

Full week-by-week history grouped by week, collapsible, with DPU trend arrows

Summary Cards

Trucks last week, defect-free count, DPU with previous-week comparison

Edit / Delete Defects

Supervisors can add, edit, or remove defect breakdown rows on existing work orders

Morning Huddle Mode

A guided full-screen presentation built into the dashboard for daily stand-up meetings.

Phase 1	Numbers	Phase 2	Issue Review	Phase 3	New Business
	- Last week OEE, Availability, Performance, Quality - Trucks completed vs. target - DPU vs. quarterly goal - Rework hours and percentage - Month-to-date and year-to-date averages - Trend arrows vs. previous week and month		- Steps through all open issues one by one - Shows foreman, category, description, age - Add update or mark solved inline - Skip issues to return to later - Supervisor dropdown for who resolved - Updates sync to dashboard in real time		- Grid of all foremen on one screen - One-tap to open issue entry form - Part or process issue type selection - Category dropdown per issue type - Submits directly to dashboard - Confirmation shown after submit

API Design — 40+ Endpoints

Issues

POST /issues

GET /issues

PUT /issues/{id}

DELETE /issues/{id}

POST /issues/bulk

GET /issues/summary/counts

POST /issues/{id}/updates

Work Orders

POST /work-orders

GET /work-orders

DELETE /work-orders/{id}

PUT /work-orders/{id}/read

POST /work-orders/{id}/defects

GET /work-orders/{id}/defects

DELETE /work-orders/{id}/defects/{id}

OEE

GET /oee/summary

GET /oee/goals

PUT /oee/goals

GET /oee/goal-history

POST /oee/goal-history

PUT /oee/goal-history/{id}

DELETE /oee/goal-history/{id}

Config

GET /truck-types

POST /truck-types

DELETE /truck-types/{id}

GET /defect-types

POST /defect-types

GET /foremen

POST /foremen

Labor

POST /indirect-labor

GET /indirect-labor

DELETE /indirect-labor/{id}

POST /rework

GET /rework

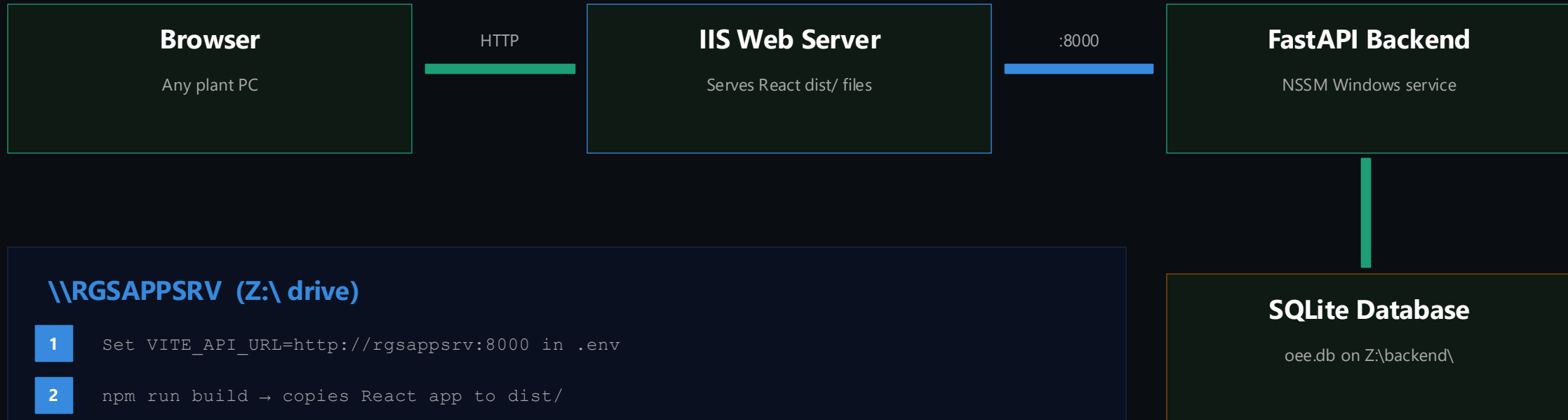
Auth / Export

POST /auth/verify

POST /auth/change-password

GET /export/all

Deployment Architecture



\\RGSAPPSRV (Z:\ drive)

- 1 Set `VITE_API_URL=http://rgsappsrv:8000` in `.env`
- 2 `npm run build` → copies React app to `dist/`
- 3 `xcopy dist\* Z:\frontend\dist\ /E /Y`
- 4 `copy main.py, database.py` → `Z:\backend\`
- 5 `python fix_db.py` on server to run migrations
- 6 Push restart button → `OEEBackend` service restarts
- 7 Test at `http://rgsappsrv`

Key Engineering Challenges

Duplicate Defect Entries

Problem

Adding a defect to a work order created 10–20 duplicate rows in the database

Fix

Isolated to cascading `onSaved()` callbacks triggering multiple re-renders. Fixed by removing `onSaved` from defect operations and using direct state updates.

total_defects Drift

Problem

The work order defect count kept getting out of sync with the actual breakdown rows

Fix

Used `db.flush()` before summing — ensures the new defect is in the session before calculating the total, replacing the old add-on-top logic.

Page Multiplication Bug

Problem

The entire dashboard rendered multiple times on scroll — page content repeated infinitely

Fix

`SupervisorManagePanel.jsx` had been overwritten with the full dashboard code. Restored it to a simple supervisor management modal.

OEE Calculation Consistency

Problem

Last week numbers, monthly averages, and huddle mode showed different values for the same week

Fix

Centralized the week metric formula into a single `computeWeekMetrics()` function used by `OEEMetrics`, `MorningHuddle`, and the summary endpoint.

What Was Built

12

Database Tables

40+

API Endpoints

20+

React Components

3

User Roles

4

OEE Metric Components

3

Huddle Phases

✓ Replaced Excel morning huddle

✓ Live issue tracking & notifications

✓ Automated OEE calculation

✓ Deployed to plant network

Thank You

OOE Dashboard · Manufacturing Intelligence System

Stack

FastAPI · React · SQLite · Recharts · NSSM · IIS

`http://rgsappsrv`

GitHub

`github.com/1-rich04/oeo-project`

Private repository